

Fixed-Length Paths II

Time limit: 1.00 s

Memory limit: 512 MB

Given a tree of n nodes, your task is to count the number of distinct paths that have at least k_1 and at most k_2 edges.

Input

The first input line contains three integers n , k_1 and k_2 : the number of nodes and the path lengths. The nodes are numbered $1, 2, \dots, n$.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output

Print one integer: the number of paths.

Constraints

- $1 \leq k_1 \leq k_2 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 2 3 1 2 2 3 3 4 3 5	6